FEDERICO GAMBASSI

AI Engineer — LLM Systems & Backend (Python · Rust)

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AI/Backend engineer with 4+ years building and operating production AI systems end-to-end. Designs LLM agents and evaluation pipelines with industrial rigor (frozen eval sets, regression gates, red-teaming), and ships the backends underneath them in Rust and Python. Founder of Dymora, a live smart-building SaaS with paying customers — solo-built from Rust API and Stripe billing to device-fleet automation and production monitoring.

WORK EXPERIENCE

Founder — Dymora (smart-building SaaS) Nov 2025 – Present

- Solo-built and operate a production smart-home platform (Rust/axum, Postgres, Redis): 37-endpoint API with OpenAPI spec, 800+ automated tests, hexagonal architecture, subscription lifecycle state machine — live at dymora.eu with paying customers.
- Built production Stripe billing (subscriptions, Klarna, customer portal) with two-layer webhook idempotency (Redis + Postgres transactional dedup) and signature verification; zero billing incidents since go-live.
- Run the full operation: staged CI/CD (auto-deploy staging, gated prod promotions), 6 VPSs across separated staging/prod environments, zero-touch device provisioning (Ubuntu autoinstall, LUKS+TPM2), self-hosted VPN control plane, append-only encrypted backups, independent dead-man-switch monitoring.

AI Engineer — TomatoAI Feb 2026 – Present

- Architected TOM, a production LLM agent (planner → executor → synthesizer) that answers Italian natural-language questions against a restaurant-management platform; provider-agnostic tool-calling layer over OpenAI and Anthropic APIs with strict-schema hardening for both vendors.
- Built the evaluation infrastructure as a first-class system: 17k-LOC gate-structured eval suite (larger than the agent's source), red-team suite, latency/cost instrumentation — regressions block release.
- Delivered an LLM invoice-classification system with research-grade methodology: frozen eval sets, blind-audited human label-noise ceiling (91–96% by axis, n=200), precision-first metric design (F0.5 with recall guardrails), reproducibility checks that fail on metric drift.
- Built a shift-scheduling optimizer on OR-Tools CP-SAT (16 pluggable constraint modules, each independently tested) with SARIMAX demand forecasting, deployed as three AWS Lambdas (SAM, staging/prod separation, 75% coverage gate enforced in pre-commit).

Cofounder/CTO — StableBitcoin Sept 2025 – Jan 2026

- Designed an OTC swap protocol for BTC ↔ risk-regularized token (sBTC): Solana programs in Rust/Anchor with Pyth oracle pricing, collateral-ratio solvency enforcement, and on-chain fee logic; devnet integration test suite in TypeScript.
- Built backend-authorized SPL reward claims with ed25519 signature verification and replay protection; Node.js backend for wallet-facing swap UI.

AI Engineer — Mathema SRL May 2023 – Jan 2026

- Architected an asynchronous AI inference pipeline for time-series data (Python/PyTorch).
- Improved transformer model accuracy by 4–5% using RL-based fine-tuning techniques.
- Built ArtSeal, a Rust (axum) system for on-chain artwork certification on Solana with a Next.js frontend.

Data Scientist — Cluster Reply May 2022 – May 2023

- Deployed autonomous ETL pipelines on Azure monitoring supply-chain inventory for a multinational coffee company, cutting processing time 30%.
- Led Azure/Power BI courses for a US multinational, mentoring 50+ students.

SELECTED PROJECTS

- **comando** (github.com/comprido96/comando) — LoRA fine-tune of Qwen3-4B (MLX, Apple Silicon) for Italian smart-home NLU, beating a prompted gpt-5-mini baseline 98.5% vs 90.25% exact-match on a frozen hand-verified eval set at $0 inference cost; published with data-size ablation and failure taxonomy.
- **LetsDAT-Otc-Swap** (github.com/comprido96) — Solana/Anchor OTC swap: Pyth oracle integration, minimum-collateral checks, BPS fees, overflow-checked release profile, devnet integration tests.
- **HomeEdge-Orchestrator** (github.com/comprido96) — 5-crate Rust workspace for edge orchestration: agent/controller split, typed shared crate, Dockerized, architecture-documented.
- **Model-Based Offline Deep RL** (M.Sc. thesis) — COMBO-style model-based offline RL applied to hybrid-vehicle energy management.

TECHNOLOGIES

Languages: Python, Rust, TypeScript/JavaScript, SQL

LLM/AI: OpenAI & Anthropic APIs (tool calling, strict schemas), eval engineering (frozen sets, gates, red-teaming), LoRA fine-tuning (MLX/mlx-lm), RAG (faiss, sentence-transformers), PyTorch, OR-Tools CP-SAT, SARIMAX

Backend: axum, tokio, sqlx, FastAPI, Node.js, Postgres, Redis, Stripe

Infra: Docker, GitHub Actions CI/CD, Ansible, AWS (Lambda/SAM), Azure, Linux

Blockchain: Solana/Anchor, Pyth

EDUCATION & CERTIFICATIONS

Let's Get Rusty Accelerator (Rust) Jan–Mar 2026

M.Sc. Mathematical Engineering — Politecnico di Torino 2019–2022

B.Sc. Mathematics — University of Firenze 2015–2019